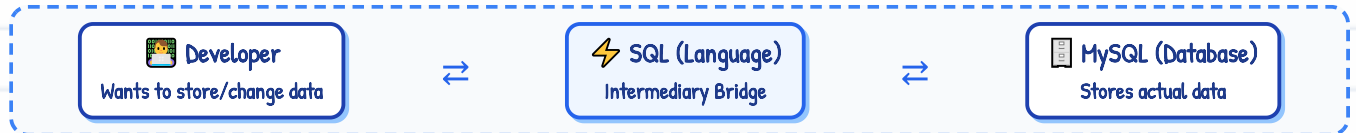


TOPIC 01: DATABASE & SQL FUNDAMENTALS

Developer, SQL & MySQL Ka Relationship (Flowchart):

Database ke saath ek developer kaise communicate karta hai, uska clear architecture:



What is a Database? (Database Kya Hai?)

Database **data ka ek organized collection** hota hai jise is tarah structure aur format kiya jata hai ki data ko **aasaan se access**, manage, modify aur retrieve kiya ja sake.

Why Database? (Database Kyu Use Karte Hain?)

- **Large Data Storage:** Gigabytes aur Terabytes of data bina slow huye store kar sakta hai.
- **High Security & Permissions:** User authentication aur role-based access provide karta hai.
- **Scalability & Performance:** Indexing aur optimization se millions of rows me se 1 millisecond me data dhoondh nikalta hai.
- **Easy CRUD Operations:** Data ko Insert karna, Update karna, ya Delete karna bohot aasaan aur safe hota hai.
- **Data Consistency (ACID Properties):** System crash hone par bhi data corrupt nahi hota.

 **Memory Hook:** MySQL ek Database Engine (Software) hai jo data store karta hai, jabki SQL ek Standard Query Language hai jiske zariye hum MySQL se baatein karte hain!

TOPIC 02: SQL vs NoSQL (THE MASTER COMPARISON)

Relational (SQL) vs Non-Relational (NoSQL):

Feature / Parameter	SQL (Relational DB)	NoSQL (Non-Relational DB)
Data Structure	Data Tables (Rows & Columns) me store hota hai.	Documents (JSON), Key-Value, ya Graphs me store hota hai.
Schema	Fixed / Strict Schema: Pehle columns aur data types define karne hote hain.	Dynamic / Flexible Schema: Har document me alag fields ho sakti hain.
Scaling	Vertical Scaling: Server me RAM/CPU badhana padta hai.	Horizontal Scaling: Naye saste servers add karke scale hota hai (Sharding).
Relations & Joins	Table relationships aur complex JOIN queries support karta hai.	JOINS support nahi hote ya bohot limited hote hain.
Best For	Complex queries, banking, e-commerce transactions (ACID).	Big data, real-time analytics, rapid prototyping, social feeds.
Popular Examples	MySQL, PostgreSQL, Oracle, SQLite, SQL Server	MongoDB, Cassandra, Redis, Neo4j, CouchDB



Interview Tip: SQL me Table ko **Relation**, Rows ko **Tuples**, aur Columns ko **Attributes** kaha jaata hai!

TOPIC 03: RDBMS DESIGN & TOPIC 04: DATABASE QUERIES

Topic 03: Relational Table Design (Schema Terminology)

- **Columns (Attributes / Design / Schema):** Table ka structure define karte hain (jaise: id, name, email, marks).
- **Rows (Tuples / Records):** Ek single individual entity ka actual data (e.g. (101, 'Shivam', 'shivam@gmail.com', 95)).

Topic 04: Database Management Queries (CRUD on Database)

Database create karne, delete karne aur inspect karne ke standard commands:

```
database-queries.sql

-- 1. Create a new Database
CREATE DATABASE college;
CREATE DATABASE IF NOT EXISTS college; -- Best Practice (No Error if exists)

-- 2. View all existing databases
SHOW DATABASES;

-- 3. Select / Switch to database to work inside it
USE college;

-- 4. View all tables inside current database
SHOW TABLES;

-- 5. Delete / Remove Database (Permanent destruction!)
DROP DATABASE college;
DROP DATABASE IF EXISTS college;
```



Caution: DROP DATABASE chalane se database aur uske andar ke saare tables aur data permanently delete ho jaate hain!

TOPIC 05: CREATE TABLE & MYSQL DATA TYPES

Create Table Syntax:

```
CREATE TABLE student (  
  id INT PRIMARY KEY,  
  name VARCHAR(50) NOT NULL,  
  age INT CHECK (age ≥ 18),  
  city VARCHAR(30) DEFAULT 'Delhi'  
);
```

create-table.sql

Common MySQL Data Types:

- **INT / INTEGER:** Standard numbers (e.g. roll_no INT).
- **TINYINT:** Chhote integers (-128 to 127), boolean ke liye bhi use hota hai (0 ya 1).
- **CHAR(n):** Fixed-length string (e.g. CHAR(10) me 'abc' likhne par bhi 10 memory slots occupy honge). Fast execution.
- **VARCHAR(n):** Variable-length string (e.g. VARCHAR(50) me 'abc' likhne par sirf 3 characters ki memory use hogi). Memory efficient!
- **BLOB (Binary Large Object):** Images, audio, videos ya binary files store karne ke liye.
- **DATE & DATETIME:** Date store karne ke liye (YYYY-MM-DD) aur Time (YYYY-MM-DD HH:MM:SS).



CHAR vs VARCHAR: Agar string ki length hamesha fixed ho (jaise PIN code ya Phone number), toh CHAR use karo. Agar variable length ho (jaise Name, Email), toh VARCHAR use karo!

TOPIC 06: SQL CONSTRAINTS (DATA INTEGRITY RULES)

SQL Ke 6 Sabse Important Constraints:

Constraints wo rules hote hain jo table ke columns par lagaye jaate hain taaki invalid data enter na ho sake:

- **1. NOT NULL:** Column kabhi bhi khali (null) nahi ho sakta.
`name VARCHAR(50) NOT NULL`
- **2. UNIQUE:** Column ki sabhi values alag (unique) honi chahiye, duplicate values allow nahi hongii.
`email VARCHAR(100) UNIQUE`
- **3. DEFAULT:** Agar user koi value provide nahi karta, toh automatic default value set ho jaati hai.
`city VARCHAR(30) DEFAULT 'Delhi', salary INT DEFAULT 25000`
- **4. CHECK:** Sirf wahi values allow karega jo specific condition ko satisfy karti hain.
`CONSTRAINT age_check CHECK (age >= 18 AND city = 'Delhi')`
- **5. PRIMARY KEY:** Table ke har row ko uniquely identify karta hai. Yeh **NOT NULL + UNIQUE** ka combination hota hai. Ek table me sirf **EK** Primary Key ho sakti hai.
`id INT PRIMARY KEY` ya `PRIMARY KEY (id)`
- **6. FOREIGN KEY:** Do tables ke beech link banata hai aur referential integrity maintain karta hai.
`FOREIGN KEY (cust_id) REFERENCES customer(id)`



Primary Key vs Foreign Key: Primary key parent table me record ko uniquely identify karti hai, aur Foreign key child table me wahi id store karke parent table se link karti hai!

TOPIC 07: INSERT & SELECT & TOPIC 08: WHERE CLAUSE

Topic 07: Data Insertion & Retrieval (Basic DML/DQL)

```
insert-and-select.sql

-- Insert single / multiple rows
INSERT INTO student (id, name, age, city)
VALUES
  (101, 'Shivam', 21, 'Delhi'),
  (102, 'Rahul', 20, 'Mumbai'),
  (103, 'Anjali', 22, 'Delhi');

-- Fetch all columns & all rows
SELECT * FROM student;

-- Fetch specific columns only
SELECT name, city FROM student;
```

Topic 08: WHERE Clause & Operators

Data ko specific conditions ke basis par filter karne ke liye WHERE clause use hota hai:

- **Arithmetic:** +, -, *, /, % (e.g. marks + 5)
- **Comparison:** =, !=, >, >=, <, <= (e.g. WHERE age >= 18)
- **Logical:**
 - AND: Dono condition true honi chahiye (WHERE age > 18 AND city = 'Delhi').
 - OR: Koi ek condition true ho (WHERE city = 'Delhi' OR city = 'Mumbai').
 - BETWEEN: Range check (WHERE marks BETWEEN 80 AND 95).
 - IN: List me se match (WHERE city IN ('Delhi', 'Pune', 'Goa')).
 - NOT: Reverse condition (WHERE city NOT IN ('Delhi')).
 - LIKE: Pattern search (WHERE name LIKE 'S%' - Starts with S).

TOPIC 09: LIMIT, ORDER BY & TOPIC 10: AGGREGATE FUNCTIONS

Topic 09: LIMIT & ORDER BY Clauses

- **LIMIT Clause:** Result me aane wale rows (tuples) ki maximum quantity set karta hai:
`SELECT * FROM student LIMIT 3;` (Top 3 rows return karega)
`SELECT * FROM student LIMIT 2, 3;` (Offset 2 se agle 3 rows - Pagination me use hota hai)
- **ORDER BY Clause:** Output ko Ascending (ASC - default) ya Descending (DESC) sort karta hai:
`SELECT * FROM student ORDER BY age ASC;`
`SELECT * FROM student ORDER BY marks DESC;`

Topic 10: Aggregate Functions in SQL

Aggregate functions multiple rows par calculations perform karke ek single summary value return karte hain:

Function	Purpose & Kya Karta Hai	Example Query
COUNT()	Total rows ya non-null values count karta hai	<code>SELECT COUNT(*) FROM student;</code>
MAX()	Column ki sabse highest / maximum value nikalta hai	<code>SELECT MAX(marks) FROM student;</code>
MIN()	Column ki sabse lowest / minimum value nikalta hai	<code>SELECT MIN(age) FROM student;</code>
SUM()	Column ki sabhi numeric values ka total jod nikalta hai	<code>SELECT SUM(salary) FROM employee;</code>
AVG()	Column ka arithmetic average (mean) nikalta hai	<code>SELECT AVG(marks) FROM student;</code>

TOPIC 11: GROUP BY, HAVING & TOPIC 12: CLAUSE ORDER

Topic 11: GROUP BY & HAVING Clauses

- **GROUP BY Clause:** Same values wale rows ko group karke summary rows banata hai. Iske saath hamesha Aggregate functions use hote hain:
`SELECT city, COUNT(id) FROM student GROUP BY city;`
(Yeh har city ke total students count karke batayega!)
- **HAVING Clause:** Groups par condition lagane ke liye use hota hai.
Kyu WHERE nahi use karte? Kyuki WHERE individual rows par chalta hai aur aggregate functions (jaise `COUNT(*) > 2`) ke saath kaam nahi kar sakta!
`SELECT city, COUNT(id) FROM student GROUP BY city HAVING COUNT(id) > 2;`

Topic 12: General Order of SQL Clauses (Blueprint)

SQL query likhne ka standard order aur MySQL engine ka internal execution order:



Execution Order: Likhte waqt SELECT pehle aata hai, lekin Database sabse pehle FROM (table dhoondhta hai), phir WHERE (filter karta hai), phir GROUP BY, phir HAVING, aur sabke baad SELECT execute karta hai!

TOPIC 13: UPDATE, DELETE & TOPIC 14: ALTER TABLE

Topic 13: Update & Delete Existing Rows (DML)

```
update-and-delete.sql

-- 1. UPDATE: Modify data in existing rows
UPDATE student
SET city = 'Pune', marks = 92
WHERE id = 102;

-- 2. DELETE: Remove existing rows from table
DELETE FROM student
WHERE marks < 33; -- Always include WHERE condition!
```

Topic 14: ALTER TABLE (Change Table Schema / Structure)

Table ka structure (columns, datatypes, constraints) modify karne ke commands:

- **Add Column:** ALTER TABLE student ADD COLUMN grade VARCHAR(2);
- **Drop Column:** ALTER TABLE student DROP COLUMN grade;
- **Rename Table:** ALTER TABLE student RENAME TO college_students;
- **Change Column (Rename Column + Datatype):**
ALTER TABLE student CHANGE COLUMN age student_age INT NOT NULL;
- **Modify Column (Change Datatype / Constraint only):**
ALTER TABLE student MODIFY student_age TINYINT DEFAULT 18;

TOPIC 15: TRUNCATE vs DROP vs DELETE (THE COMPARISON)

The Ultimate Confusion Cleared (Exam Favorite):

Feature	DELETE	TRUNCATE	DROP
Command Type	DML (Data Manipulation)	DDL (Data Definition)	DDL (Data Definition)
What It Deletes	Sirf specific rows (based on WHERE clause).	Saara data / rows delete karta hai.	Data + Table structure dono delete kar deta hai.
Table Structure	Table structure intact rehta hai.	Table structure intact rehta hai.	Table structure database se gayab (destroy) ho jata hai!
WHERE Clause	Allowed (WHERE id = 101)	NOT allowed (Sabhi rows delete hoti hain)	NOT allowed
Speed & Logs	Slow (har row ko individually log karta hai).	Super Fast (table ko deallocate karta hai).	Instant (drop table completely).

Complete Working SQL Script (VS Code Style):

```
complete-master-queries.sql

CREATE DATABASE IF NOT EXISTS school_db;
USE school_db;

CREATE TABLE students (
  id INT PRIMARY KEY,
  name VARCHAR(50) NOT NULL,
  marks INT CHECK (marks ≥ 0 AND marks ≤ 100),
  city VARCHAR(30) DEFAULT 'Delhi'
);

INSERT INTO students VALUES
(1, 'Aman', 85, 'Delhi'), (2, 'Shradha', 95, 'Mumbai'), (3, 'Karan', 72, 'Delhi');

-- Group by City with Having & Order By
SELECT city, AVG(marks) AS avg_marks FROM students
GROUP BY city HAVING AVG(marks) > 75
ORDER BY avg_marks DESC;
```